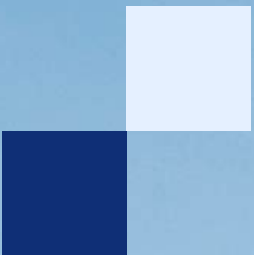


REAL LLM JUDGE ENSEMBLE FOR ADVERSARIAL ESSAY GENERATION

**KAGGLE COMPETITION: LLMS – YOU CAN'T PLEASE
THEM ALL**



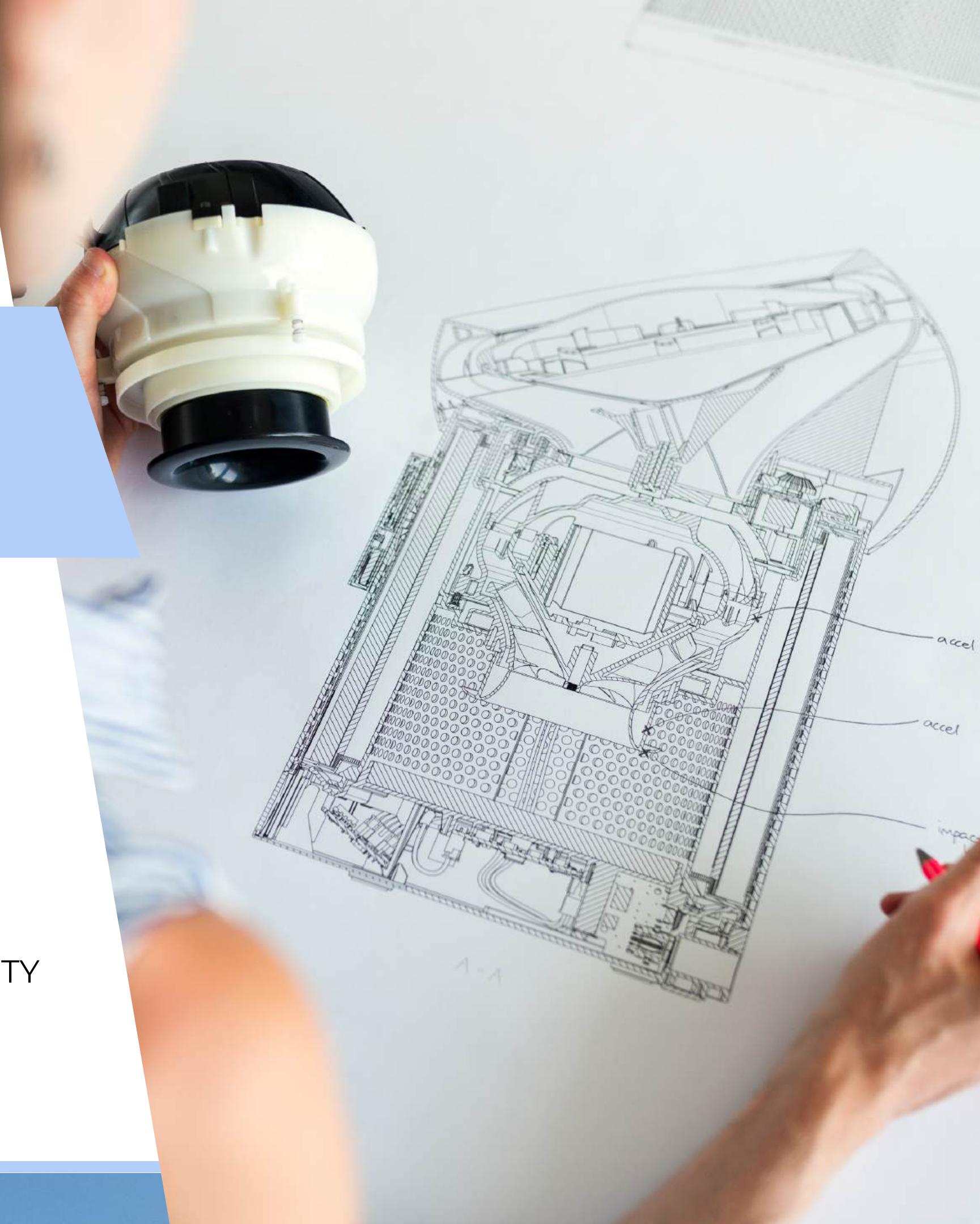
PRESENTED BY

GRUOP AFIQQ

INTRODUCTION

Project Overview

- Investigates robustness of LLM-as-a-Judge systems
- Different LLM judges may evaluate essays differently
- Goal: generate essays that maximize judge disagreement
- MAINTAIN RELEVANCE AND ACCEPTABLE WRITING QUALITY



PROBLEM UNDERSTANDING

KAGGLE PROVIDES ESSAY
TOPICS

MULTIPLE HIDDEN LLM JUDGES
EVALUATE SUBMISSIONS

NEED ESSAYS THAT RECEIVE
DIVERSE EVALUATIONS

FINAL SCORE DEPENDS ON
JUDGE DISAGREEMENT

BASELINE METHOD

Generate one balanced academic
essay

SIMPLE AND REPRODUCIBLE

FIXED PROMPT TEMPLATE

LIMITATION

- HIGH AGREEMENT AMONG JUDGES
- LOW DIVERSITY IN EVALUATION SCORES

SINGLE WRITING STYLE

PROPOSED METHOD

Multi-Strategy Essay Generation

Generate multiple candidate essays:

- Academic Balanced
- Creative Reflective
- Concise Direct
- Skeptical Logic
- Pro/Con Contrast
- Structured Headings
- Empathetic Social
- Uncertainty Aware



REAL LLM JUDGE ENSEMBLE

JUDGE PERSONAS

- STRICT_ACADEMIC
- CREATIVITY_FRIENDLY
- CONCISION_FRIENDLY
- SKEPTICAL_LOGIC

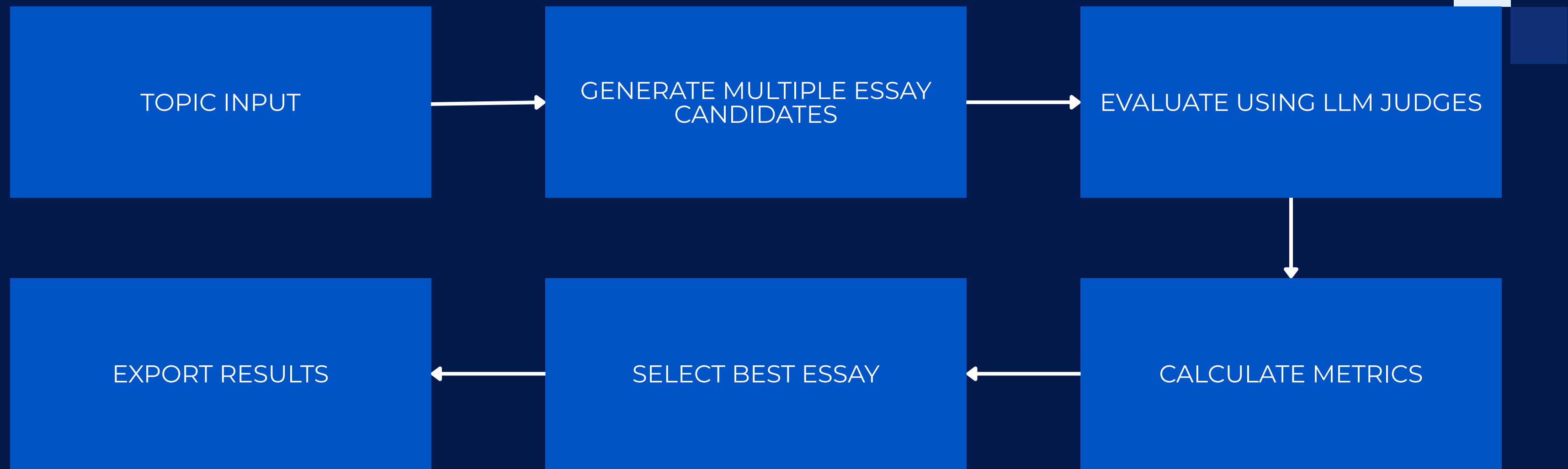
IMPLEMENTATION

- GEMINI API
- OLLAMA (LOCAL LLM)
 - LLAMA3.1
 - QWEN3
 - MISTRAL
 - GEMMA3

IMPROVEMENT

- REPLACES HANDCRAFTED/RANDOM SCORING
- PROVIDES REALISTIC EVALUATION

SYSTEM WORKFLOW & SELECTION OBJECTIVE



SELECTION METRICS

- MEAN SCORE
- STANDARD DEVIATION
- SCORE RANGE
- WORD COUNT

IMPROVEMENTS MADE

SYSTEM ENHANCEMENT

1. REAL AI-BASED EVALUATION

- REPLACED RANDOM/SIMULATED JUDGE SCORES
- INTRODUCED MULTIPLE JUDGE PERSONAS:
- STRICT JUDGE
- CREATIVE JUDGE
- LOGICAL JUDGE
- IMPROVED EVALUATION REALISM AND CREDIBILITY

2. EXPANDED EXPERIMENT COVERAGE

- ADDED MULTIPLE TOPICS:
 - HEALTHCARE POLICY
 - SELF-RELIANCE & ADAPTABILITY
 - AI IN EDUCATION
 - CLIMATE CHANGE
 - ADDED MULTIPLE ESSAY-GENERATION STRATEGIES

3. AUTOMATED EXPERIMENT PIPELINE

- AUTOMATED GENERATION, EVALUATION, AND SELECTION PROCESS
- TESTED MULTIPLE TOPICS × STRATEGIES × JUDGES
- IMPROVED REPRODUCIBILITY AND SCALABILITY

4. ADVANCED ANALYSIS & DOCUMENTATION

- ADDED STATISTICAL METRICS:
 - MEAN SCORE
 - STANDARD DEVIATION
 - SCORE RANGE
- EXPORTED RESULTS TO CSV FOR TRACKING AND ANALYSIS

IMPACT ON PROJECT QUALITY

- HIGHER TECHNICAL DEPTH
- STRONGER EXPERIMENTAL RIGOR
- BETTER RESULTS & ANALYSIS
- IMPROVED DOCUMENTATION
- MORE PROFESSIONAL PRESENTATION

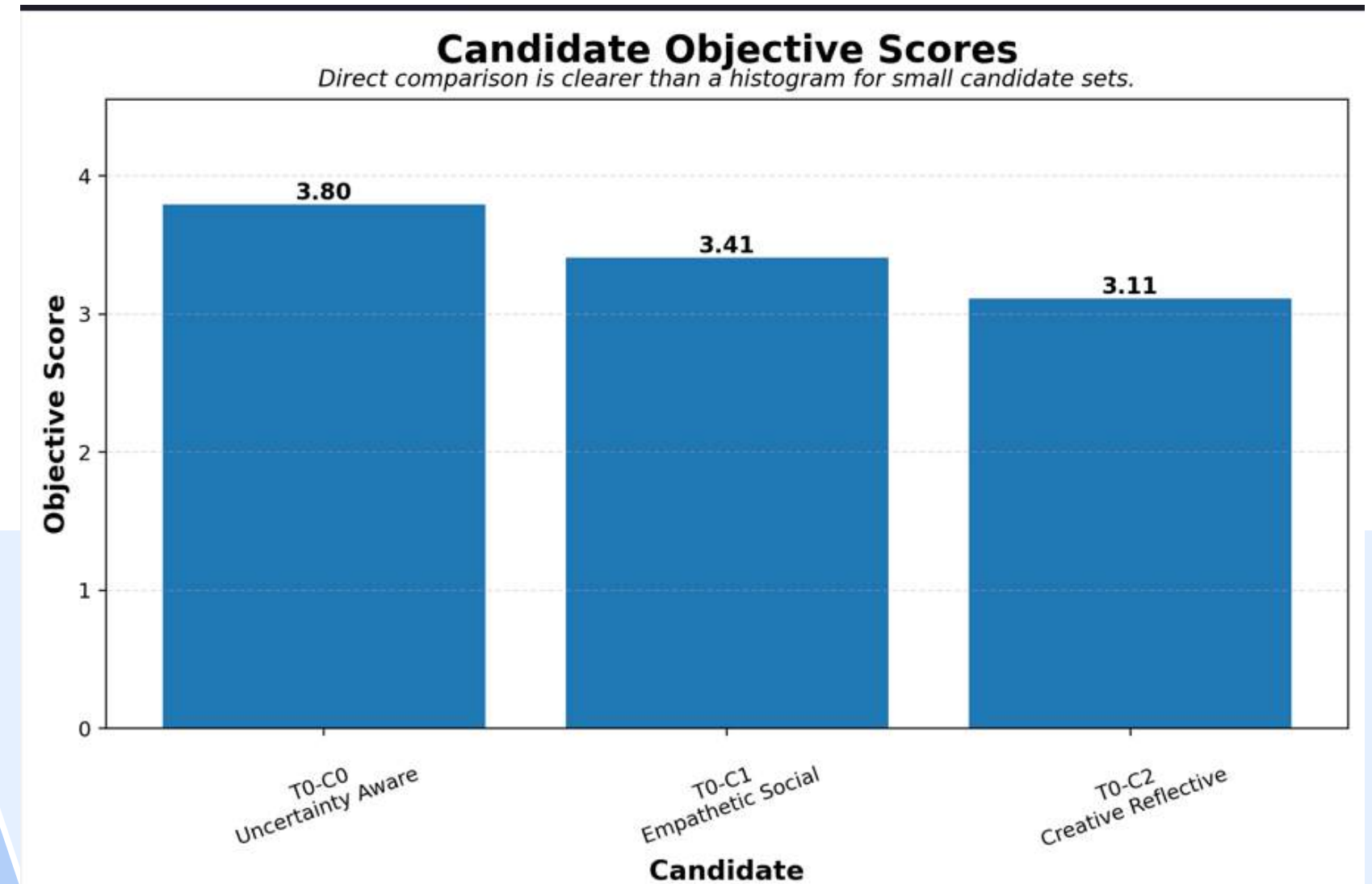
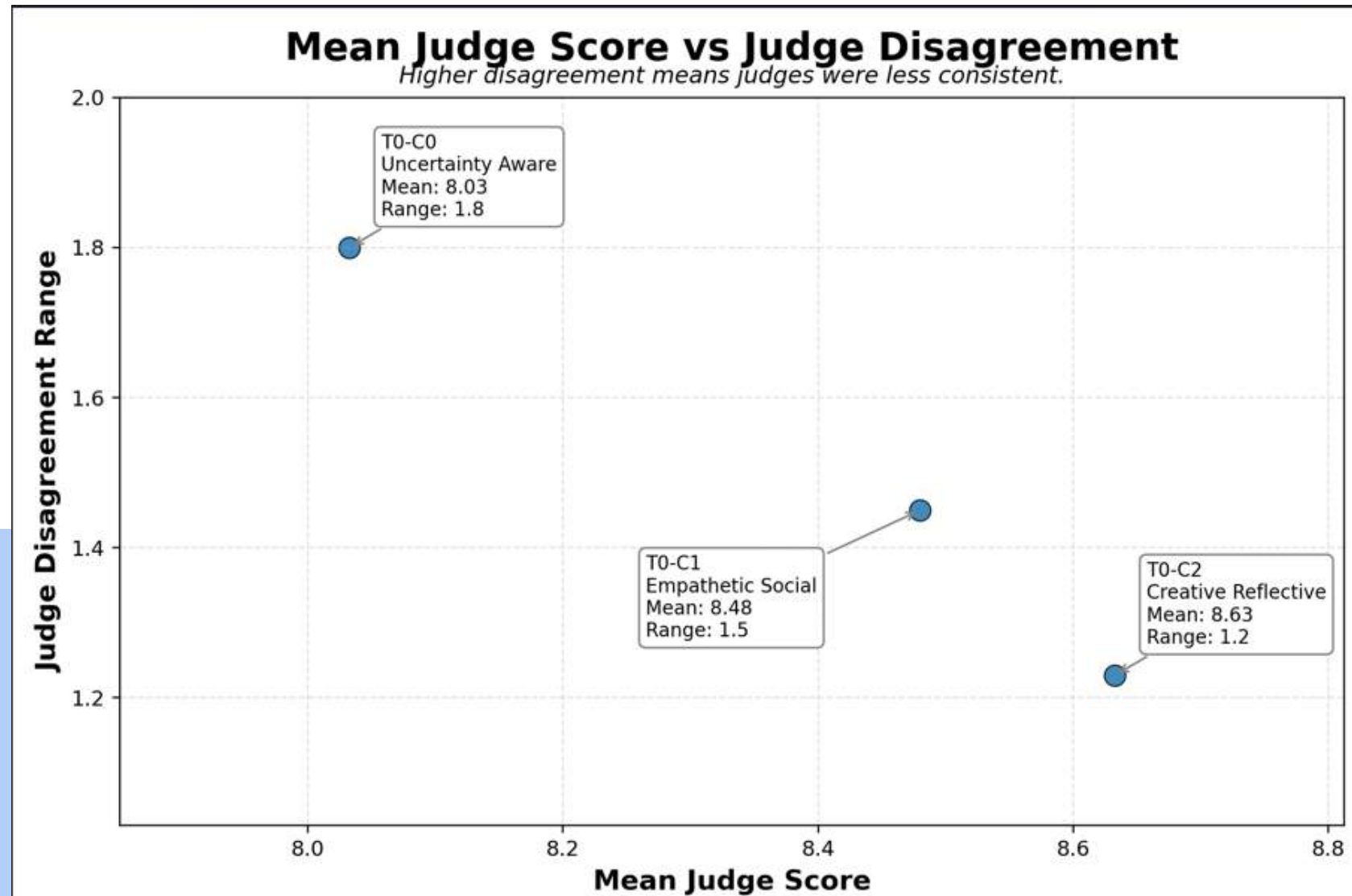
RESULTS, ANALYSIS & TECHNICAL CONTRIBUTIONS

Data

id	topic						
0	Is technology making students more independent or more dependent?						
1	Should universities use AI tools to grade essays?						
2	Can social media improve public discussion?						C

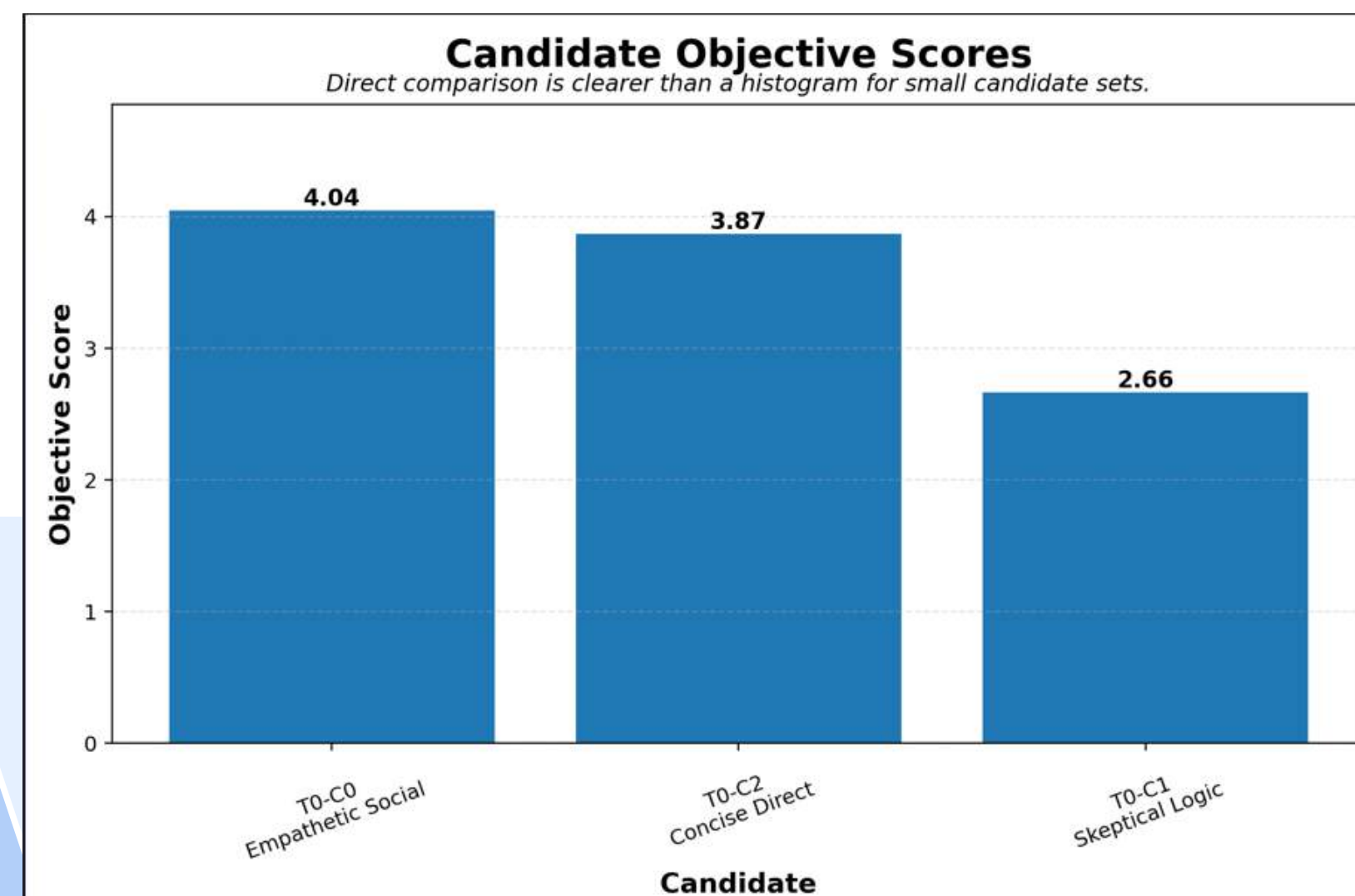
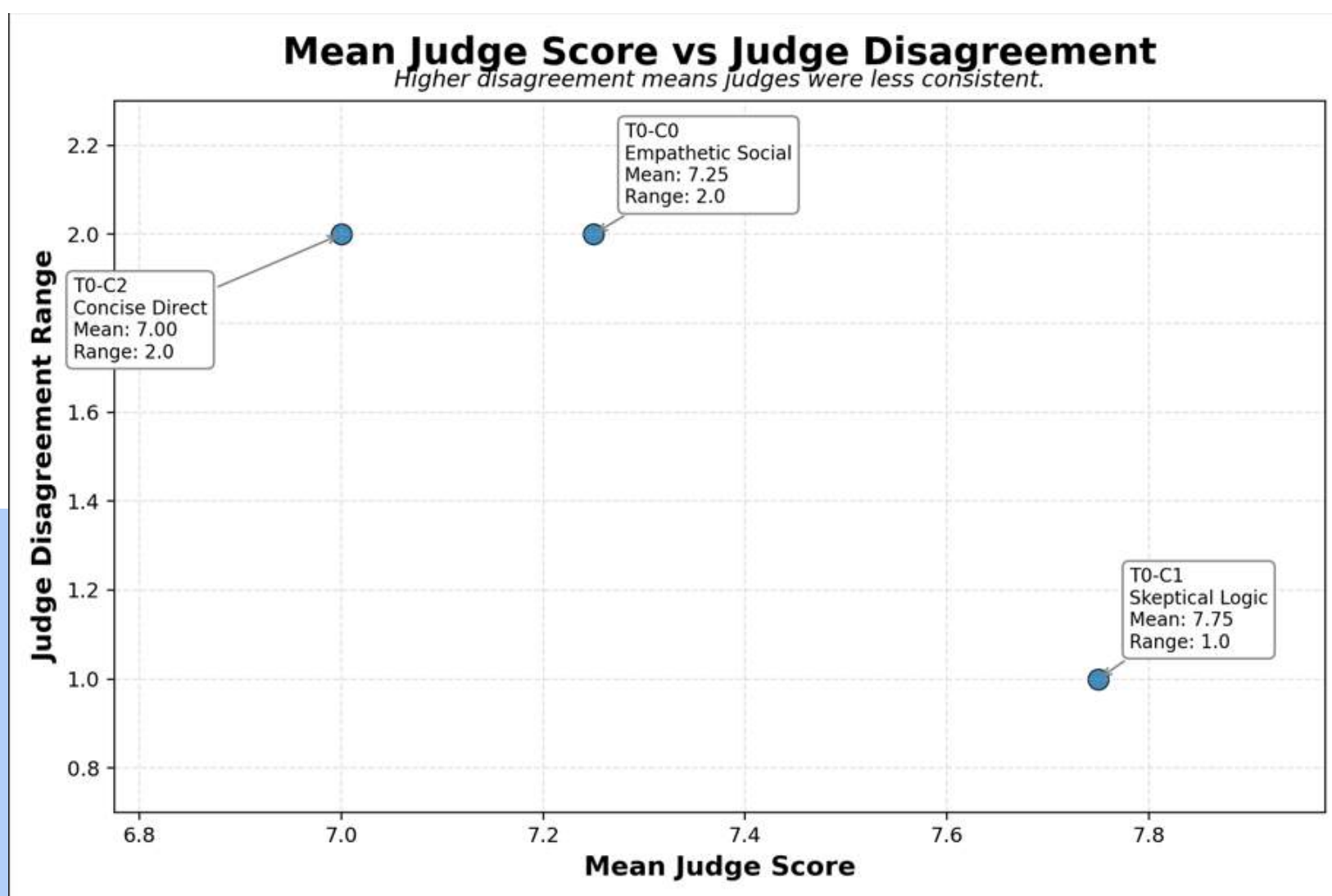
RESULTS, ANALYSIS & TECHNICAL CONTRIBUTIONS

Heuristic model



RESULTS, ANALYSIS & TECHNICAL CONTRIBUTIONS

ollama qwen model

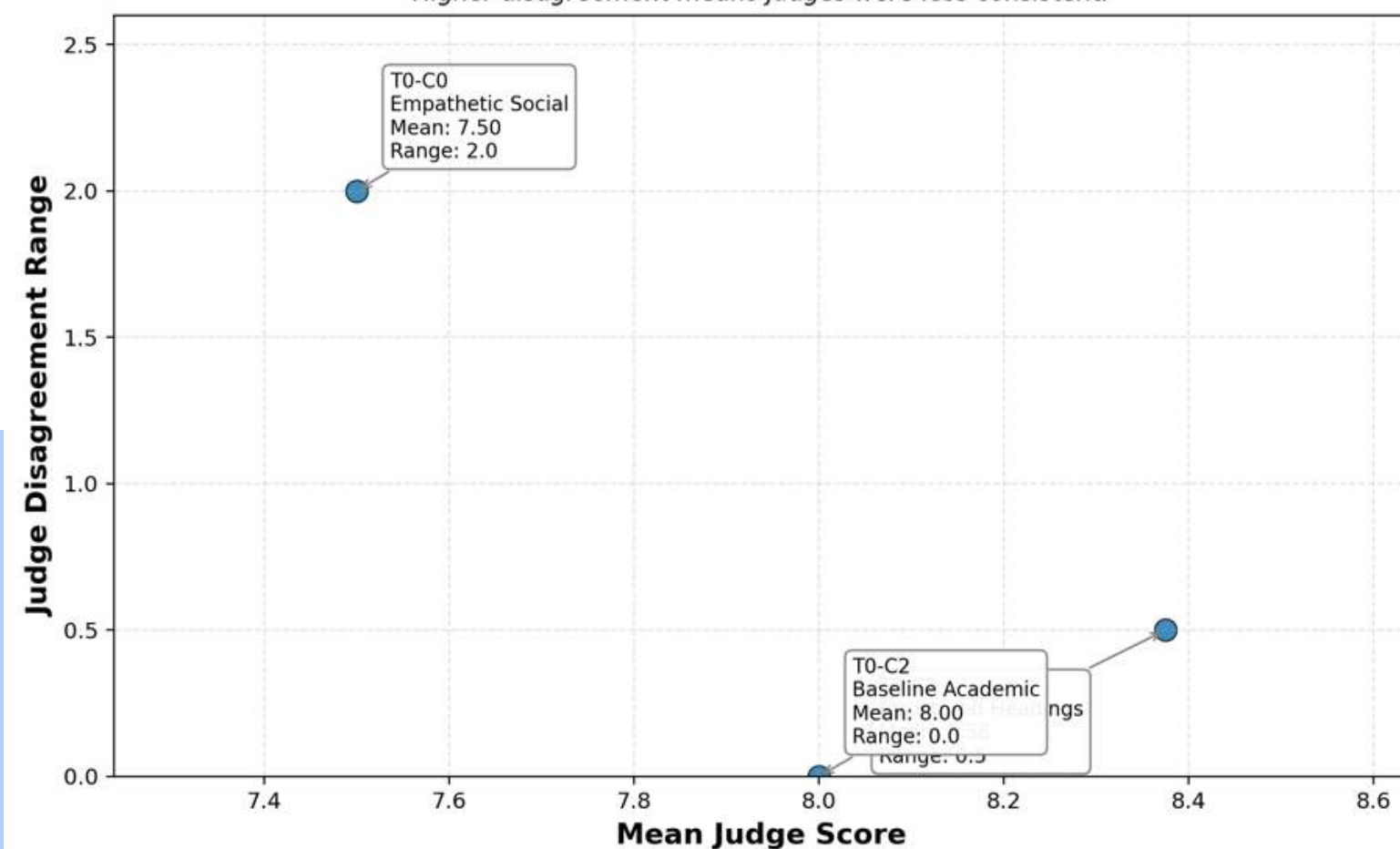


RESULTS, ANALYSIS & TECHNICAL CONTRIBUTIONS

ollama mistral model

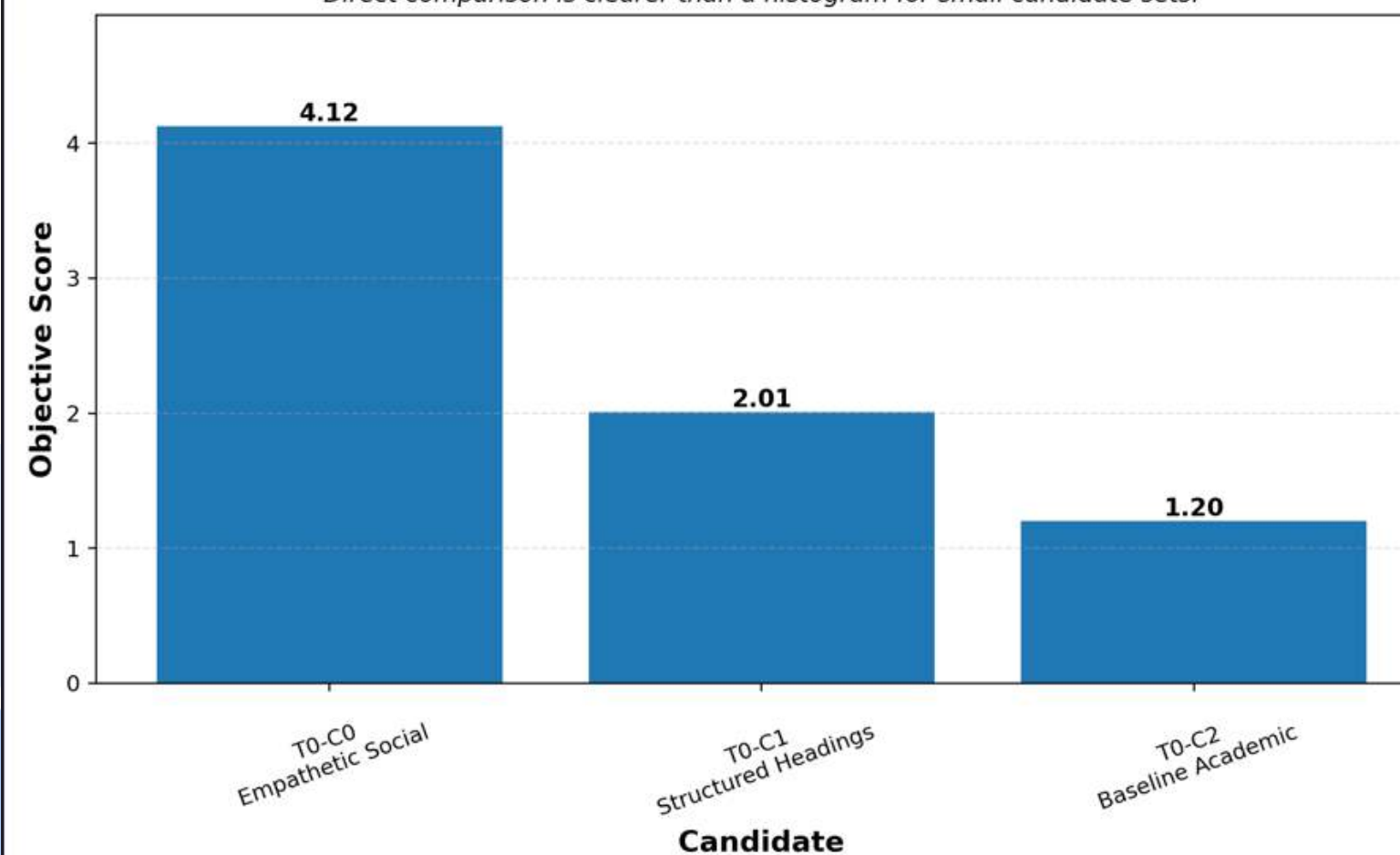
Mean Judge Score vs Judge Disagreement

Higher disagreement means judges were less consistent.



Candidate Objective Scores

Direct comparison is clearer than a histogram for small candidate sets.

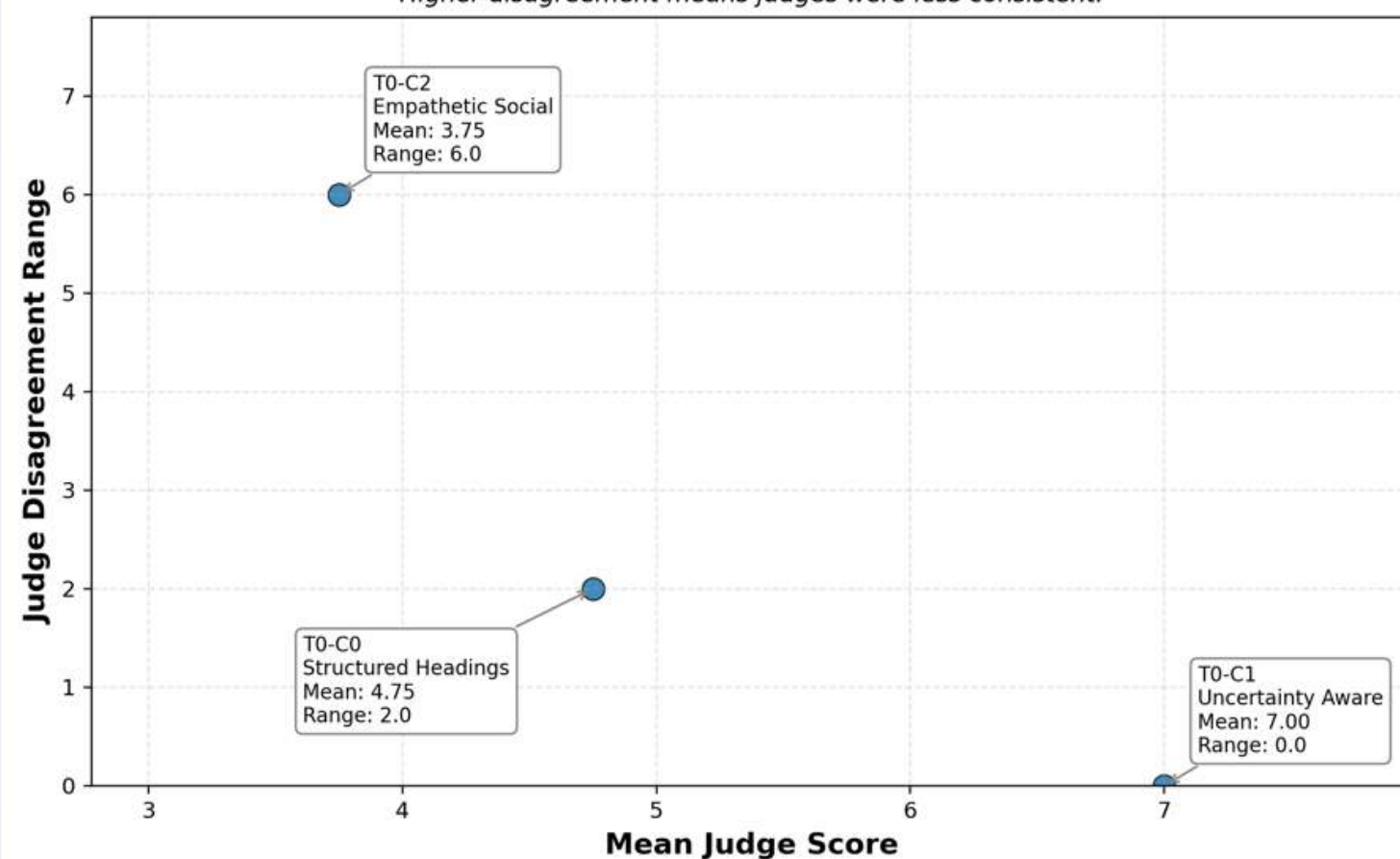


RESULTS, ANALYSIS & TECHNICAL CONTRIBUTIONS

gemini model

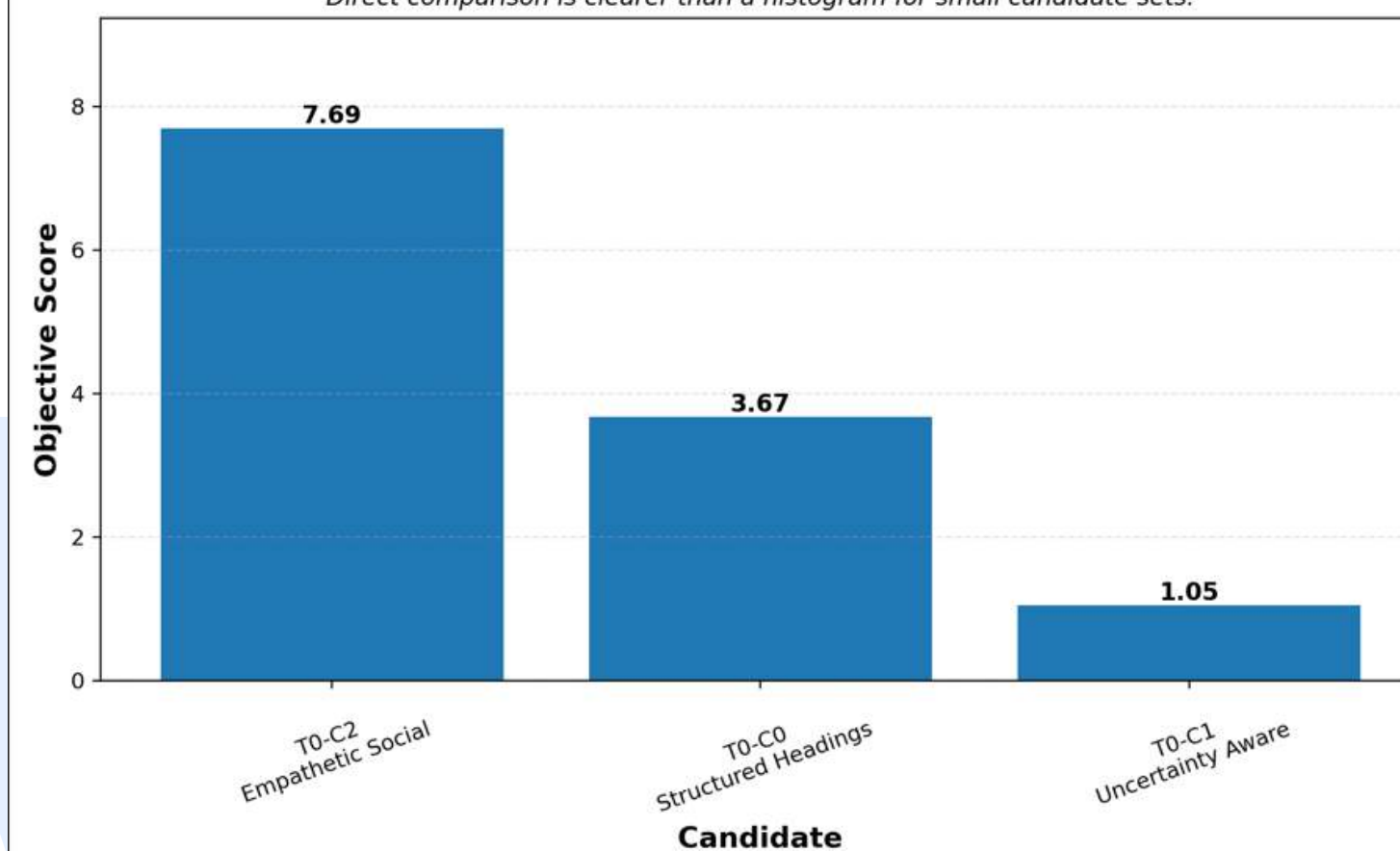
Mean Judge Score vs Judge Disagreement

Higher disagreement means judges were less consistent.



Candidate Objective Scores

Direct comparison is clearer than a histogram for small candidate sets.



LIMITATIONS, CONCLUSION & FUTURE WORK

Limitations

- Hidden Kaggle judges unavailable
- Gemini/Ollama only approximate actual judges
- LLM outputs may vary
- API cost considerations

Future Work

- Add more judge models
- Improve scoring objectives
- Test larger topic datasets
- Develop visualization dashboard

Conclusion

- Developed a complete LLM evaluation workflow
- Generated diverse essay candidates
- Replaced heuristic scoring with real LLM judges
- Improved experimental rigor and reproducibility

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THANK YOU.

